PWCC: Probability-Weighted Combinatorial Coverage for Risk-Informed Scenario Testing of Autonomous Systems

Borhaneddine Hamadou
Université Paris-Saclay, CEA List
Palaiseau, France
borhaneddine.hamadou@cea.fr

Chokri Mraidha
Université Paris-Saclay, CEA List
Palaiseau, France
chokri.mraidha@cea.fr

Nikolaos Matragkas
Université Paris-Saclay, CEA List
Palaiseau, France
nikolaos.matragkas@cea.fr

Abstract

Scenario-based testing of autonomous systems faces a fundamental unsolved problem: knowing when to stop. Existing coverage metrics measure either structural diversity or combinatorial breadth, but neither tells you how much of the system's expected real-world operation remains untested. A test suite can score well on every existing metric while systematically missing the conditions the system will actually encounter.

We introduce Probability-Weighted Combinatorial Coverage (PWCC), a metric that simultaneously measures coverage of the Operational Design Domain (ODD) and test adequacy. PWCC learns a probabilistic model of expected operating conditions from operational profiling data, uses it to assign probability mass to t-way combinations of scenario features, and declares testing adequate when the uncovered mass falls below a risk-calibrated threshold. This gives practitioners something no existing metric provides: a principled, quantitative answer to "have we tested enough?"

Evaluated on four autonomous driving datasets against seven baselines, PWCC demonstrates three properties. First, it is the strongest predictor of fault detection where the operational distribution carries signal, reaching Spearman $r_s = +0.981$ on DeepCollision — 0.167 above the best combinatorial baseline at a lower interaction order. Second, it suppresses false sufficiency: on DeepScenario, PWCC achieves FSR = UFC = 0 across all splits versus CC's mean UFC $\geq 95.8\%$, false sufficiency eliminated entirely, and, judged at its own stopping point, surfaces 99.6% of failure-triggering combinations where combinatorial saturation stops after 6.4% of the campaign; on failure-rich datasets UFC drops by up to 4.9×. Third, it is stable: the score varies by at most 0.006 across a ten-fold increase in Monte Carlo sample size and grows monotonically with suite size.

These results show that operational probability weighting is the decisive factor in avoiding dangerous false sufficiency verdicts; and that PWCC transforms coverage from a descriptive statistic into a risk-informed stopping decision.

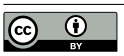
CCS Concepts

• **Software and its engineering** → **Software testing and debugging**; *Software safety*; • **Computer systems organization** → *Embedded and cyber-physical systems.*

Keywords

test adequacy, operational design domain, scenario-based testing, combinatorial interaction testing, safety assurance, autonomous driving, SOTIF

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1 Introduction

Safety-critical autonomous systems must operate safely across an effectively unbounded space of environmental conditions. Validating such systems through exhaustive testing is impossible. Scenario-based testing has emerged as the primary answer to this challenge [42], offering a structured way to exercise system behavior under representative conditions. Yet despite significant progress in scenario generation [5], selection [28], and prioritization [4], a fundamental question remains unanswered: **when have we tested enough?**

This question has direct safety and certification consequences. Safety certification frameworks such as ISO 21448 (SOTIF) [17] require not only evidence that tests were executed, but a justified claim that additional testing would not substantially increase coverage of safety-relevant conditions. Without a principled stopping criterion, test campaigns face two symmetric failures: premature termination that leaves dangerous conditions unexplored, or indefinite continuation that wastes resources without meaningful safety gain. Neither outcome is acceptable in a certification context.

The coverage gap. Current adequacy metrics for scenario-based testing fall into two categories, each with a critical limitation. White-box structural metrics, such as neuron coverage [37], require access to model internals and do not reliably predict fault detection [44]; they are not our focus here. Among black-box approaches, diversity-based metrics assess the geometric spread of test cases in feature space [1, 10, 33]. They provide useful descriptive insight into how broadly the scenario space has been explored, but tell you nothing about how much of the ODD has actually been covered, and offer no principled basis for deciding when testing is sufficient. Combinatorial coverage metrics [24] measure the fraction of t-way parameter combinations exercised by a test suite. They are efficient and scalable, but treat every combination as equally important, a critical flaw when some environmental conditions are orders of magnitude more likely than others in operation.

The missing dimension. None of these approaches connects coverage to the expected operational frequency of the conditions being tested. A test suite can achieve high combinatorial coverage while systematically neglecting the most common driving conditions, or high diversity while concentrating effort on operationally irrelevant edge cases. What is missing is a coverage metric grounded in the operational distribution of the system under test — one that measures not just *what* has been tested, but *how much of what the system will actually encounter* has been adequately exercised.

Operational profiles have long guided testing in software reliability engineering toward the most frequently executed code paths [31]. Risk-based statistical testing [50] and reliability growth models [38] establish that test adequacy should be assessed against operational likelihood, not structural completeness. These principles are well established in traditional software testing but have not been operationalized for scenario-based testing of autonomous systems, where the input space is continuous, high-dimensional, and governed by complex environmental dependencies.

This paper. We introduce **Probability-Weighted Combinatorial Coverage (PWCC)**, a metric that fills this gap. PWCC learns a probabilistic model of expected operating conditions from operational profiling data, assigns probability mass to t-way combinations of scenario variables, and declares testing adequate when the uncovered mass falls below a risk-calibrated threshold. This simultaneously addresses two needs: it measures Operational Design Domain (ODD) coverage in operationally meaningful units, and it provides a principled, quantitative stopping criterion grounded in explicit confidence and risk parameters.

Research question. Current scenario coverage metrics measure what has been tested, not how much of what the system will encounter in operation has been adequately exercised. This gap makes it impossible to justify stopping a test campaign on safety grounds. We therefore ask:

> *How can we quantify the fraction of expected real-world operation that remains insufficiently tested, and use that quantity as a principled, risk-informed criterion for test campaign termination?*

Contributions. (1) **PWCC**, a probability-weighted combinatorial coverage metric that weights t-way combinations by credited operational probability mass from a learned ODD model (Section 3). (2) A **principled stopping criterion** parameterised by a per-combination risk budget ρ, confidence level $1{-}\alpha$, and coverage threshold ε (Section 3.5). (3) An **empirical evaluation** on four autonomous driving datasets against seven baselines across five research questions (Sections 4, 5).

2 Background and Related Work

Test adequacy, coverage metrics, and diversity. We position PWCC along three axes — its *goal* (a risk-informed adequacy verdict and stopping rule), its *inputs* (black-box tabular scenarios plus operational profiling data), and its *output* (a probability-weighted coverage score with an explicit residual-mass guarantee) — and review prior work along them. Test adequacy criteria define when a suite is sufficient [49]. Structural criteria (statement, branch, mutation) [2] assume source access, and neural coverage metrics [21, 29, 37] do not reliably predict fault detection [14, 44]; neither transfers to black-box scenario testing. Diversity-based black-box metrics measure geometric spread via Euclidean distance [28], normalised compression distance [1, 10], or Instance Space Analysis [33]. TISA [33] is the closest prior work: it characterises diversity through convex-hull area and boundary coverage, but does not weight test cases by operational likelihood, imposes no minimum exposure criterion, and offers no principled stopping rule. We adopt TISA as our diversity baseline; its original evaluation already benchmarks it against Euclidean and NCD diversity [1, 10, 28], so we take it as representative of that family. A second family measures the physical or semantic space a suite *experiences*: PhysCov [16] reduces sensor reachability, S3C [45] abstracts scene graphs, and STARS [40] builds coverage from ODD-level scene abstractions. All quantify space *already exercised* rather than operation left *insufficiently tested*, and require vehicle state, dynamics, or sensor streams (e.g., LiDAR/radar) that PWCC's tabular pipeline does not assume; parameter coverage [25] is white-box, and behaviour-and-ODD coverage [7] needs human-authored taxonomies. PWCC assumes none of these.

Combinatorial interaction testing, ODD modelling, and weighted extensions. Combinatorial Interaction Testing (CIT) exercises all t-way parameter combinations to detect interaction faults [24, 34] and has been applied to ADS scenario selection [5, 22]. Classical CIT weights every combination uniformly regardless of operational frequency — a critical flaw for open-world systems — and weighted extensions [8, 9] use engineering judgment or fault history rather than a learned model. PWCC instead derives weights directly from operational profiling data; to our knowledge, it is the first to connect combinatorial coverage to the expected real-world distribution. The Operational Design Domain (ODD) defines where a system is designed to operate safely [17, 23, 30]; most work treats it as a manually specified boundary. Data-driven ODD characterisation [20] and learning monitorable ODDs from behaviour [43] share PWCC's empirical-grounding perspective; PWCC extends it by making the learned probabilistic ODD model the direct foundation of the coverage metric and stopping criterion.

Statistical testing, operational profiles, and stopping criteria. Operational profiles have long guided testing toward frequent paths [31]; risk-based statistical testing [50] and reliability growth models [38] extend this to resource allocation and residual reliability. Risk-based acceptance frameworks on exposure accounting and operational profiles [13] give scenario-level tolerability thresholds; PWCC's residual-mass guarantee is a coverage-oriented counterpart, usable alongside them in a safety case. In the driving domain, naturalistic-and-adversarial environments [11] and importance sampling [18] accelerate discovery of rare critical events, and critical-scenario search is surveyed in [48]; these target unbiased *risk estimation* (how likely a failure is), whereas PWCC produces an *adequacy verdict* over t-way combinations (how much operation has been exercised) — PWCC's statistical-testing lineage, and it can consume importance-sampled suites. Confidence-based stopping for t-way covering arrays [24, 34] declares completeness at combinatorial saturation regardless of frequency; PWCC replaces this with a continuous criterion calibrated by a risk budget and confidence level — to our knowledge, the first stopping criterion for scenario-based

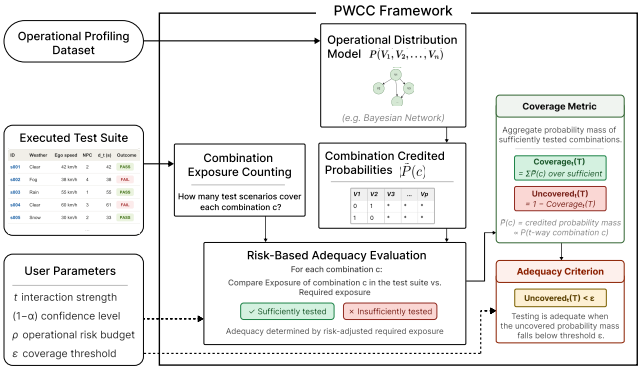


Figure 1: Overview of the PWCC framework: profiling data and the executed test suite flow through operational modelling and risk-based adequacy evaluation to produce a coverage score and stopping verdict.

testing grounded in operational likelihood, statistically justified, and interpretable as a residual coverage bound. Scenario-based testing spans search-based [12, 28] and LLM-based [6, 46] generation, simulation [27], and prioritisation [4, 47], yet when a suite is sufficient has received little attention.

Conformal prediction. Conformal prediction (CP) yields prediction sets with distribution-free, finite-sample coverage under exchangeability [39], with Monte-Carlo variants [3] tightening intervals in safety-critical settings. CP and PWCC certify different objects: CP covers a model's *output predictions* under exchangeability, whereas PWCC issues an adequacy verdict over *input-space regions* under an operational profile, assuming no exchangeability on test order. They are compatible — CP-style intervals are a natural vehicle for quantifying uncertainty in PWCC's coverage estimates near its adequacy threshold (Section 6).

3 Approach

PWCC quantifies how much of the expected operational variation of a system under test has been adequately exercised by a finite test suite. The framework has four components: (1) a probabilistic model of expected operating conditions learned from operational profiling data, (2) a probability-weighted t-way coverage metric over scenario variables, (3) a risk-calibrated minimum exposure criterion, and (4) a principled adequacy criterion that serves as a stopping rule. Figure 1 gives an overview.

3.1 Scope and Assumptions

PWCC operates on *concrete* test scenarios, each a fully specified assignment of values to n scenario variables $\mathbf{V} = \{V_1, \ldots, V_n\}$ representing observable environmental and situational parameters (e.g., weather, traffic density, road geometry). It requires no access to system internals and is therefore applicable to any black-box system under test. The framework assumes that the operational profiling dataset $\mathcal{D}_{\text{prof}}$ is relevant to the intended deployment conditions, but not that it is exhaustive: the probabilistic model makes residual uncertainty explicit, supporting transparent safety argumentation. Profiling data may originate from field observations, simulation campaigns, meteorological records, or expert elicitation.

3.2 Inputs

PWCC takes three inputs:

(1) **Operational profiling dataset** $\mathcal{D}_{\text{prof}}$: scenarios representative of the expected operating conditions, used to learn a probabilistic model of the ODD.
(2) **Executed test suite** $\mathcal{T}$: the set of executed scenarios, each associated with a simulated duration d_τ (in seconds), independent of execution hardware.
(3) **User-defined parameters**:
- t — interaction order: t=2 considers all pairwise combinations, higher values capture more complex interactions at increased cost.
- ρ — per-combination risk budget: the maximum tolerable critical event rate per t-way combination. $\rho = 0.01$ requires ruling out a failure rate exceeding 0.01/hour per combination.
- α — miss probability: the acceptable probability of failing to detect a systematic problem. Typical values are 0.05 (95% confidence) or 0.01 (99% confidence).
- ε — uncovered mass threshold: testing is adequate when uncovered operational probability mass falls below ε.

3.3 Probabilistic Model of Operating Conditions

Let $P(\mathbf{V}) = P(V_1, V_2, \ldots, V_n)$ denote the joint probability distribution of scenario variables under the intended operational conditions. PWCC requires a model $\hat{P}$ of this distribution, learned from $\mathcal{D}_{\text{prof}}$. The model must support marginal probability queries of the form:

$$\hat{P}(c) = \hat{P}(V_{i_1} = v_1, \ldots, V_{i_t} = v_t), \tag{1}$$

where c is a t-way combination of scenario variable values. Any probabilistic model supporting such queries is a valid instantiation of PWCC. Section 3.7 describes the specific instantiation used in this work.

A key property of the model is that it assigns non-zero probability to operationally plausible combinations that may be rare or unobserved in $\mathcal{D}_{\text{prof}}$. This enables PWCC to reason about untested but operationally relevant conditions, going beyond what purely data-driven frequency counts can express.

3.4 Probability-Weighted t-way Coverage

3.4.1 Credited Probability Mass. Let C_t denote the set of all t-way combinations over the scenario variable space. Because t-way combinations overlap — a single scenario contributes to multiple combinations — directly summing marginal probabilities $\hat{P}(c)$ across combinations leads to double counting. We therefore define a *credited probability mass* $\tilde{P}(c)$ that partitions total probability mass exactly once across all combinations.

Each scenario s induces $K(s)$ valid t-way projections, where $K(s) = \binom{n}{t}$ in the unconstrained case. The credited mass of combination c is:

$$\tilde{P}(c) = \sum_{\substack{s \in \mathcal{S} \\ s \text{ consistent with } c}} \frac{\hat{P}(s)}{K(s)}, \tag{2}$$

where the sum is over all fully specified scenarios s consistent with c. This construction distributes each scenario's probability mass equally across all its valid t-way projections, guaranteeing $\sum_{c \in C_t} \tilde{P}(c) = 1$ (Equation 3), so the credited masses form a proper probability distribution over combinations usable for risk allocation and coverage computation.

$$\sum_{c \in C_t} \tilde{P}(c) = 1. \tag{3}$$

The split is *within-scenario* only, so more probable scenarios contribute more mass and rarity is preserved (Section 6).

3.4.2 Minimum Required Exposure. We allocate the per-combination risk budget ρ across t-way combinations in proportion to their credited probability mass [31]. Combination c receives a condition-specific maximum tolerable event rate:

$$\lambda_{\max}(c) = \frac{\rho}{\tilde{P}(c)}. \tag{4}$$

This allocation bounds each combination's individual contribution to expected risk: $\tilde{P}(c) \cdot \lambda(c) \le \rho$. Note that summing over all combinations yields $\sum_c \tilde{P}(c) \cdot \lambda(c) \le \rho \cdot |C_t|$, not ρ — the allocation $\lambda_{\max}(c) = \rho / \tilde{P}(c)$ is a *proportionality criterion*, not a global risk budget.

Proposition 3.1 (Global risk bound under proportional allocation). *Under $\lambda^\star(c) = \rho \cdot \tilde{P}(c)$, the global aggregate satisfies $\sum_c \tilde{P}(c) \cdot \lambda^\star(c) = \rho \cdot \sum_c \tilde{P}(c)^2 \le \rho$, where the inequality follows from $\tilde{P}(c) \le 1$ and Equation (3).*

We model the occurrence of critical events under combination c as a homogeneous Poisson process with rate $\lambda(c)$ [38]. Under this model, the probability of observing at least one event during exposure time E is $1 - e^{-\lambda(c)E}$. To detect a systematic problem with rate $\lambda(c) \ge \lambda_{\max}(c)$ with confidence $1 - \alpha$, we require:

$$1 - e^{-\lambda_{\max}(c)\, E} \ge 1 - \alpha \implies E \ge \frac{-\ln(\alpha)}{\lambda_{\max}(c)}. \tag{5}$$

Substituting Equation 4 yields the minimum required exposure for combination c:

$$E_{\min}(c) = \frac{-\ln(\alpha) \cdot \tilde{P}(c)}{\rho}. \tag{6}$$

This formulation has an important property: operationally frequent combinations receive higher exposure requirements, concentrating testing effort where it matters most. Rare combinations require less exposure, reflecting their lower contribution to operational risk.

Remark 1. *The alternative tight-bound allocation $\lambda^\star(c) = \rho \cdot \tilde{P}(c)$ inverts the exposure scaling, producing requirements unachievable at simulation scale (e.g., ≈3,000 simulated hours for a single moderately probable combination at $\rho = 0.01$, $\alpha = 0.05$), which motivates the proportional criterion in Equation 4.*

3.4.3 Accumulated Exposure and Sufficiency. For each combination c, the total accumulated test exposure is:

$$E_c = \sum_{\tau \in \mathcal{T}(c)} \frac{d_\tau}{3600}, \tag{7}$$

where $\mathcal{T}(c) \subseteq \mathcal{T}$ is the set of executed tests consistent with c, and d_τ is the simulated duration of test τ in seconds (the factor 3600 converts to hours, matching the unit of ρ and λ). A combination is *sufficiently tested* when its accumulated exposure meets the minimum required threshold:

$$\text{Sufficient}(c, \mathcal{T}) = \begin{cases} 1 & \text{if } E_c \ge E_{\min}(c), \\ 0 & \text{otherwise.} \end{cases} \tag{8}$$

Note that this criterion is a *coverage* metric, not a *fault-absence* claim: $E_{\min}(c)$ is the exposure required to detect a systematic fault at rate $\lambda_{\max}(c)$ with confidence $1 - \alpha$ *if one exists*. Any failure that occurs during E_c is observed by construction in simulation; the stopping rule governs whether enough time has elapsed to surface a fault, not whether observed failures are absent. A suite meeting the criterion may still contain observed failures; any such failures must be addressed separately before the system is declared safe for deployment.

Worked example. To make the mechanics concrete, take t=2, ρ=0.01/h, α=0.05. A moderately frequent combination c = ⟨precip=heavy, road=wet⟩ with credited mass $\tilde{P}(c)$=0.02 tolerates rate $\lambda_{\max}(c) = \rho/\tilde{P}(c)$=0.5/h and, by Equation (6), must accrue $E_{\min}(c) = -\ln(0.05) \cdot 0.02/0.01 \approx 6.0$ h of exposure to rule out a systematic fault at 95% confidence. If the tests exercising c sum to E_c=6.5 h it is sufficiently tested (Equation (8)); a rarer ⟨precip=heavy, road=icy⟩ with $\tilde{P}$=0.002 needs only ≈0.6 h, so effort concentrates where operation is frequent.

3.4.4 Coverage Metric. The PWCC score is the total credited probability mass of all sufficiently tested combinations:

$$\text{PWCC}_t(\mathcal{T}) = \sum_{\substack{c \in C_t \\ \text{Sufficient}(c, \mathcal{T}) = 1}} \tilde{P}(c). \tag{9}$$

This answers the question: *what fraction of the expected operational variation, expressed in terms of t-way interactions, has been adequately exercised?* The complementary uncovered mass is:

$$\text{Uncovered}_t(\mathcal{T}) = 1 - \text{PWCC}_t(\mathcal{T}). \tag{10}$$

3.5 Adequacy Criterion and Stopping Rule

Testing is considered adequate when the uncovered credited probability mass falls below the threshold ε:

$$\text{Uncovered}_t(\mathcal{T}) < \varepsilon. \tag{11}$$

This criterion has a direct safety interpretation: testing is adequate when less than a fraction ε of the expected operational probability mass corresponds to insufficiently tested t-way conditions. The threshold ε should be chosen based on system criticality and safety requirements [17]. For example, $\varepsilon = 0.05$ declares adequacy when at least 95% of the operational probability mass has been sufficiently tested at the specified confidence level $1 - \alpha$ and per-combination risk budget ρ.

3.6 Global Assurance on Residual Operational Exposure

The proportional allocation $\lambda_{\max}(c) = \rho/\tilde{P}(c)$ does not, by itself, induce a tight bound on aggregate risk over all combinations. However, PWCC's adequacy criterion provides a complementary global guarantee on the *unassured* portion of the operational space.

Proposition 3.2 (Residual exposure bound). *If $\text{Uncovered}_t(\mathcal{T}) < \varepsilon$, then the total credited operational probability mass of insufficiently tested combinations is bounded by ε:*

$$\sum_{\substack{c \in C_t \\ \text{Sufficient}(c,\mathcal{T})=0}} \tilde{P}(c) < \varepsilon. \tag{12}$$

This follows directly from Equation (10), yielding a two-level assurance structure: for combinations declared sufficient, ρ and α govern the required exposure; globally, ε bounds the fraction of expected operational encounters outside this assurance regime. Concretely, at least $(1-\varepsilon)$ of expected operational encounters have been tested to the chosen confidence and risk parameters, and at most ε remain insufficiently exercised — a bound on residual *unassured operational mass* directly usable in SOTIF safety arguments.

3.7 Probabilistic Model Instantiation

PWCC requires a probabilistic model $\hat{P}$ supporting the marginal queries in Equation 1. In this work, we instantiate $\hat{P}$ as a Bayesian network (BN) [36], a directed acyclic graphical model that compactly represents the joint distribution of scenario variables through conditional independence relationships.

The BN structure is learned from $\mathcal{D}_{\text{prof}}$ using the PC algorithm [41], a constraint-based structure learning method that recovers conditional independence relationships from data. Parameters are estimated using a Bayesian estimator with a BDeu prior (equivalent sample size $ESS = 5$, a standard weakly informative choice [15]), which guarantees non-zero probability for every CPT cell, including unobserved combinations.

All continuous features are discretised into four bins using cut points at the 25th, 50th, and 75th percentiles of each feature's marginal in $\mathcal{D}_{\text{prof}}$, computed once and frozen as fixed constants applied verbatim to the test set; already-categorical features (≤ 10 distinct values) are left unbinned. Where a mass point would collapse two cut points, the affected edge is displaced so the mass keeps its own bin.

The credited mass $\tilde{P}(c)$ in Equation 2 is approximated by drawing N forward samples from the BN. Each sample s contributes $1/K(s)$ to every combination it covers — no explicit $\hat{P}(s)$ weight is applied. Operational weighting is implicit: since samples are drawn proportionally to $\hat{P}(s)$, frequent combinations naturally accumulate more mass. This is an unbiased estimator of Equation 2 by the law of large numbers, and $\sum_c \tilde{P}(c) = 1$ is preserved by construction. $N = 50{,}000$ yields stable estimates with score variation below 0.006 across a ten-fold increase in sample size (Section 5.5).

The BN is one valid instantiation of PWCC. Any model supporting marginal queries — including kernel density estimators, normalizing flows, or Gaussian mixture models — can be substituted without modifying the coverage framework. The BDeu prior ensures that $\tilde{P}(c) > 0$ for all combinations, including those unobserved in $\mathcal{D}_{\text{prof}}$; however, BN parameter uncertainty is not currently propagated into adequacy decisions. Near the ε threshold, credible intervals on $\tilde{P}(c)$ derived from the Dirichlet posteriors on CPT cells would strengthen certification arguments — we identify this as a prioritised direction for future work.

Table 1: Overview of experimental datasets. n_{prof}: profiling set size; FR: failure rate.

Dataset	Scenarios	n_{prof}	Failures	FR (%)	Features
AsFault [12]	5,638	8,538	2,543	45.1	7
DeepScenario [27]	13,565	19,966	233	1.7	10
DeepCollision [26]	8,605	8,060	5,129	59.6	18
Frost [35]	5,294	41,768	389	7.3	9

4 Evaluation

4.1 Research Questions

RQ1 — Correlation with fault detection. How strongly does PWCC correlate with failures detected versus existing metrics? A metric tracking fault detection provides a reliable signal of test suite quality [19].

RQ2 — Coverage gap detection. How effectively does PWCC identify under-tested ODD regions, and does its score on $\mathcal{D}_{\text{test}}$ predict failures in a held-out $\mathcal{D}_{\text{valid}}$? Predicting unseen failure concentration provides stronger safety evidence than correlation alone.

RQ3 — False sufficiency under ODD omission. When an ODD region is withheld, how do PWCC's *False Sufficiency Rate* (FSR) and *Uncovered Failure Coverage* (UFC) compare to baselines? Lower FSR and UFC mean fewer dangerous false adequacy verdicts in deployment.

RQ4 — Stopping behaviour. When each metric triggers its own stopping rule, how much of the campaign does it consume, and what fraction of failure-triggering combinations remains undiscovered at that point? A fixed suite size cannot reveal this; a stopping criterion must be judged at the point it declares adequacy.

RQ5 — Sensitivity and stability. How sensitive is PWCC to $(\rho, \alpha, \varepsilon, t)$, and does the score grow monotonically with suite size? A deployable metric must be stable and provide consistent feedback as testing progresses.

4.2 Datasets

We evaluate on four datasets that collectively cover a range of failure rates, feature dimensionalities, and distribution regimes. Table 1 summarises their key properties.

AsFault [12] uses search-based road network generation to stress lane-keeping software; variation is concentrated in road geometry with weather held constant. Its high failure rate (45.1%) and low-dimensional feature space stress-test PWCC near metric saturation. **DeepScenario** [27] uses a deep-learning search in CARLA maximising behavioural diversity; its near-zero failure rate (1.7%) tests PWCC's ability to withhold adequacy when failures are sparse.

DeepCollision [26] uses reinforcement learning to maximise collision probability across 18 features; its failure-biased generation produces distributional skew between profiling and testing data. **Frost** is a novel dataset introduced in this work, derived from real-world meteorological records from the Norwegian Meteorological Institute's Frost API [35] and instantiated as CARLA scenarios. Because the profiling distribution is dominated by clear-weather conditions (86.7% dry) while the test suite samples all weather levels uniformly, Frost provides a controlled stress-test of PWCC under structural profiling–testing distribution mismatch.

Profiling data. For AsFault, DeepScenario, and DeepCollision, $\mathcal{D}_{\text{prof}}$ is drawn from scenarios generated by the baseline methods in the respective original papers [12, 26, 27], ensuring the probabilistic model is learned from a distribution independent of the evaluated test suite. For Frost, $\mathcal{D}_{\text{prof}}$ consists of the meteorological records described above (sizes in Table 1).

4.3 Baselines

We compare PWCC against seven baselines spanning three categories, summarised in Table 2.

Combinatorial coverage (CC, CC-k2, CC-k3). **CC** [24] measures the fraction of t-way combinations covered at least once under uniform weight. **CC-k2** and **CC-k3** are stricter variants requiring each combination to appear in at least k=2 or k=3 distinct tests. None incorporates operational probability weighting or a risk-calibrated exposure criterion.

Empirical Frequency Coverage (EFC). EFC extends CC by weighting combinations by their empirical frequency in $\mathcal{D}_{\text{prof}}$. Its key limitations are that it assigns zero weight to operationally plausible but unobserved combinations, and imposes no minimum exposure requirement.

Test-suite Instance Space Analysis (TISA). TISA [33] applies Instance Space Analysis [32] to characterise diversity through three geometric metrics: area_{IS} (convex hull area), $\text{area}_{\text{bugs}}$ (area occupied by failing scenarios), and cov_{IS} (boundary coverage fraction). TISA offers no principled stopping rule, making it inapplicable to RQ3 and RQ4.

PWCC ablations. **PWCC-Uniform** replaces the learned credited mass $\tilde{P}(c)$ with a flat weight $1/|C_t|$, retaining all other PWCC components. **PWCC-Empirical** replaces the probabilistic model with empirical frequency counts, assigning zero mass to combinations unobserved in $\mathcal{D}_{\text{prof}}$. Together these ablations isolate the contribution of (i) the minimum exposure criterion and (ii) the learned operational probability weighting.

4.4 Experimental Setup

RQ1. For each dataset we generate 30 test suites of size 2,500 with bug rates linearly spaced across the feasible range of the dataset, following the stratified sampling strategy of Neelofar and Aleti [33]. We compute all metrics on each suite and report Spearman's r_s between metric value and failure count. We additionally report r_s between coverage on $\mathcal{D}_{\text{test}}$ and failures in a held-out $\mathcal{D}_{\text{valid}}$, assessing predictive power beyond the test suite itself.

Table 2: Baseline metrics and RQ inclusion. Exp.req.: risk-calibrated minimum exposure criterion applied. ✓ = included; — = excluded. CC-k2/k3 and the three TISA metrics (area_{IS}, $\text{area}_{\text{bugs}}$, cov_{IS}) each share identical inclusion.

Metric	Weighting	Exp.req.	RQ1	RQ2	RQ3	RQ4	RQ5
PWCC	Learned BN	Yes	✓	✓	✓	✓	✓
PWCC-Uniform	Uniform	Yes	—	✓	✓	—	—
PWCC-Empirical	Empirical	Yes	—	✓	✓	—	—
CC [24]	Uniform	No	✓	✓	✓	✓	—
CC-k2/k3 [24]	Uniform	No	—	✓	✓	✓	—
EFC	Empirical	No	✓	✓	—	✓	—
TISA [33]	Geometric	No	✓	✓	—	—	✓‡

‡ TISA enters RQ5 only as a runtime comparison (Table 8); it is not part of the sensitivity analysis.

RQ2 and RQ3. Both questions share the same protocol. For each dataset and $t \in \{2, 3, 4\}$, we construct ten independent ODD omission splits. Each split fixes $k = t{+}1$ feature values simultaneously and withholds *every* scenario matching all of them; these form the held-out region $\mathcal{D}_{\text{valid}}$, and the remainder is subsampled to $|\mathcal{D}_{\text{test}}| = 2{,}500$ (disjoint from $\mathcal{D}_{\text{valid}}$). Because the region is pinned by $t{+}1$ jointly fixed values whereas a t-way metric inspects only t features at a time, no combinatorial metric can flag the omission through combination membership — detecting it requires exposure weighting (TISA's geometric sensitivity is unconstrained). For instance, at t=2 a split might fix k=3 values jointly (e.g. ⟨precip=heavy, road=wet, manoeuvre=turning⟩): every scenario matching all three is withheld, yet no pairwise metric sees the region as a unit. For each split we record the failure count in $\mathcal{D}_{\text{valid}}$ and the credited mass $\tilde{P}_{\text{omit}}$ of the omitted region. AsFault supports $t \leq 3$ only.

RQ2 reports r_s between each metric's score on $\mathcal{D}_{\text{test}}$ and the failure count in $\mathcal{D}_{\text{valid}}$ across ten splits per $(dataset, t)$; a significant negative r_s is the expected direction. RQ3 derives a binary adequacy verdict from each metric: for PWCC, $\text{Uncovered}_t(\mathcal{T}) < \varepsilon$; for unweighted baselines, combinatorial saturation (at least once, or k times for CC-k variants). FSR is the proportion of falsely declared-sufficient scenarios containing failures; UFC is the proportion of all unseen failures concealed by false verdicts. Both are lower-is-better.

RQ4. We evaluate each metric as a stopping rule. For each dataset and $t \in \{2, 3, 4\}$ we grow the suite in 1% increments over ten random orderings and trigger each metric's own stop, recording two quantities at that point: *Used*, the fraction of the campaign consumed, and *Cap*, the fraction of distinct failure-triggering t-way combinations already surfaced. PWCC stops at the first step with $\text{Uncovered}_t(\mathcal{T}) < \varepsilon$; combinatorial baselines stop at coverage plateau ($\geq 0.95\times$ their maximum over the run, since a monotonically growing suite need not reach exact saturation).

RQ5. We conduct a one-at-a-time (OAT) sensitivity analysis varying each parameter $(\rho, \alpha, \varepsilon, N)$ while holding the others at their baseline values ($\rho = 0.01$, $\alpha = 0.05$, $\varepsilon = 0.05$, $N = 50{,}000$). For each configuration we draw ten random subsamples at suite sizes from 500 to 5,000 in increments of 500 and compute the PWCC score and adequacy rate. Monotone growth is assessed visually and by the coefficient of variation across repetitions at each size.

4.5 Operational Distribution Modelling

The probabilistic model $\hat{P}$ is instantiated as a Bayesian network learned from $\mathcal{D}_{\text{prof}}$ (Section 3.7). The credited probability mass $\tilde{P}(c)$ is estimated by drawing $N = 50{,}000$ forward samples from the BN. Baseline parameters are set to $\rho = 0.01$ events/hour, $\alpha = 0.05$, $\varepsilon = 0.05$, and $t = 2$ for all experiments unless otherwise stated.

5 Results

5.1 RQ1: Correlation with Fault Detection

Figure 2 and Table 3 report Spearman r_s between each metric's score and the fault count across 30 stratified test suites per dataset. AsFault's 7-feature space supports up to $t = 3$; $t = 4$ results are therefore excluded for this dataset. Results expose a sharp, dataset-dependent pattern: PWCC is the strongest fault-detection predictor when the operational distribution *carries exploitable signal* — i.e. profiling mass concentrates on the combinations where failures actually occur — and produces no spurious correlation when it does not.

DeepCollision. PWCC dominates all baselines on the dataset with the richest operational structure (18 features, 59.6% failure rate, RL-biased generation). PWCC_{t2} reaches $r_s = +0.981$ — the highest value recorded across all metrics and all datasets — and PWCC_{t3} ($r_s = +0.925$) and PWCC_{t4} ($r_s = +0.884$) both exceed every combinatorial baseline at the same interaction order. The best CC variant, CC_{t3} ($r_s = +0.814$), trails PWCC_{t2} by 0.167 at a *lower* interaction order, confirming that probability weighting — not combinatorial breadth — drives the gain. EFC_{t4} ($r_s = +0.922$) is the closest competitor, yet still falls 5.9 points behind PWCC_{t2} and requires a higher interaction order. TISA metrics are substantially weaker (cov_{IS}: +0.610; area_{IS}: +0.374), confirming that geometric diversity is an insufficient proxy for operational fault exposure.

AsFault. AsFault operates in a near-saturated coverage regime: PWCC_{t2} scores span the narrow band $[0.973, 0.986]$ across all 30 suites, yielding $r_s = -0.484$ ($p = 0.007$). The mild negative correlation is a saturation artefact — PWCC already certifies adequate coverage for almost every suite, leaving insufficient variance to track fault rate, whereas CC and EFC continue to grow with fault rate ($r_s \in [0.828, 0.887]$, all $p < 0.001$). This is consistent with PWCC's design: it is not intended to differentiate fault rates *within* an already-adequate regime, but to detect and prevent false adequacy verdicts before that regime is reached.

DeepScenario. With only 1.7% failures and a near-uniform plateau for all metrics, no metric achieves a statistically significant correlation ($|r_s| \leq 0.252$, all $p > 0.18$). PWCC_{t2} scores are effectively zero across all suites (range: $[0,\ 1.3 \times 10^{-5}]$), correctly signalling that no suite accumulates sufficient exposure under the risk budget $\rho = 0.01$. When failures are too sparse to carry distributional signal, no coverage metric can serve as a reliable fault predictor; PWCC's behaviour is consistent with this fundamental limit.

Frost. Frost imposes a structural profiling–testing mismatch: the profiling distribution is dominated by dry, high-friction conditions (86.7% no precipitation, 70.8% friction = 0.8), while the validation set samples all weather levels uniformly. No metric achieves meaningful correlation at $t = 2$ or $t = 3$. At $t = 4$, PWCC_{t4} ($r_s = -0.517$, $p = 0.003$), CC_{t4} ($r_s = -0.562$, $p = 0.001$), and EFC_{t4} ($r_s = -0.457$, $p = 0.011$) all yield moderate negative correlations, as suites with higher fault rates concentrate on adversarial low-friction conditions assigned low probability mass by the profiling model — a structurally coherent outcome under distributional mismatch, not a metric failure.

Table 3: RQ1: Spearman r_s between metric score and fault count (30 test suites per dataset, n=2,500 each). †: $p > 0.05$; bold: highest $|r_s|$ per dataset; —: t exceeds feature-space dimensionality.

Metric	AsFault	DeepColl.	DeepScen.	Frost
PWCC_{t2}	−0.484	**+0.981**	−0.185†	−0.214†
PWCC_{t3}	−0.214†	+0.925	+0.171†	−0.301†
PWCC_{t4}	—	+0.884	+0.120†	−0.517
CC_{t2}	+0.846	+0.724	+0.008†	−0.091†
CC_{t3}	+0.887	+0.814	−0.007†	−0.014†
CC_{t4}	—	+0.830	+0.068†	−0.562
EFC_{t2}	+0.831	+0.753	−0.063†	−0.073†
EFC_{t3}	+0.828	+0.870	−0.120†	−0.127†
EFC_{t4}	—	+0.922	−0.161†	−0.457
cov_{IS}	−0.580	+0.610	−0.252†	−0.217†
area_{IS}	+0.122†	+0.374	+0.099†	+0.084†
area_{bugs}	**−0.963**	−0.880	+0.082†	+0.089†

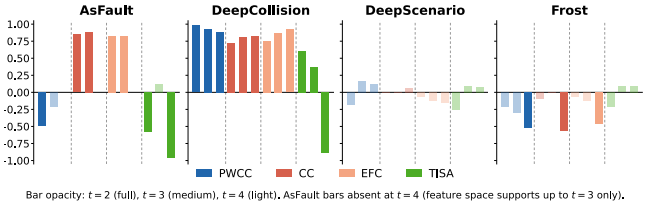


Figure 2: Spearman r_s between metric score and fault count (30 suites per dataset, n=2,500). Blue: PWCC; red: CC; orange: EFC; green: TISA. Opacity decreases with $t \in \{2, 3, 4\}$. AsFault absent at t=4 (supports $t \leq 3$ only).

RQ1

PWCC is the strongest predictor of fault detection when the operational distribution carries signal, reaching $r_s = +0.981$ on DeepCollision — 0.167 above the best combinatorial baseline at a lower interaction order (t=2 vs. t=3). On saturated or near-zero-failure datasets, PWCC correctly abstains. On Frost, moderate t=4 negative correlations are a structurally coherent consequence of distributional mismatch, confirming that probability weighting is the decisive factor where the distribution carries signal.

5.2 RQ2: Coverage Gap Detection

RQ2 asks whether a metric's score on $\mathcal{D}_{\text{test}}$ predicts the failure concentration in a held-out ODD region $\mathcal{D}_{\text{valid}}$. Each dataset is partitioned ten times by withholding a k-way combination of feature values; Spearman r_s between each metric's score and the failure count in $\mathcal{D}_{\text{valid}}$ is reported across repetitions (Table 4).

The role of $\tilde{P}$. Before examining PWCC directly, a key diagnostic is the credited probability mass of the omitted region, $\tilde{P}_{\text{omit}}$. On DeepCollision it is the strongest predictor at every interaction order

(r_s = +0.927*, +0.770*, +0.806* at t=2, 3, 4), and on DeepScenario it achieves r_s = +0.683* and +0.799* at t=2 and t=3. This directly validates PWCC's core design: the probability mass assigned to an ODD region by the Bayesian network is an accurate proxy for its failure density. Regions with higher $\tilde{P}_{\text{omit}}$ concentrate more failures, and consequently produce lower PWCC scores — which is the intended alarm.

DeepCollision. PWCC correctly signals coverage gaps on the most failure-rich dataset. PWCC_{t2} ($r_s = -0.830^*$) and PWCC_{t3} ($r_s = -0.661^*$) are both significant: a lower PWCC score on $\mathcal{D}_{\text{test}}$ reliably predicts more failures in the withheld region. The negative sign is the *expected* and *correct* direction — lower coverage signals a dangerous gap. The ablation PWCC-uniform (r_s = +0.818* at t=2) produces the *opposite* direction: it issues high coverage scores precisely when the omitted region is failure-dense. This is false sufficiency — the metric certifies adequacy where the system is most exposed. Removing operational probability weighting is the direct cause; uniform weighting cannot distinguish operationally important from unimportant regions. PWCC-empirical ($r_s = -0.867^*$ at t=2) is competitive at pairwise order but degrades at t=4, where sparse profiling data renders empirical frequency counts unreliable. cov_{IS} reaches $r_s = -0.806^*$ at t=3, matching PWCC in direction but providing no exposure threshold, no risk budget, and no stopping criterion. CC and EFC show no significant signal at any interaction order.

DeepScenario. With only 1.7% failures, PWCC operates near its resolution floor. At t=3, $\tilde{P}_{\text{omit}}$ (r_s = +0.799*) and PWCC-empirical (r_s = +0.701*) are the only significant predictors. PWCC_{t3} (r_s = +0.543, $p = 0.105$) and PWCC_{t4} ($r_s = -0.671^*$) capture the gap at higher interaction orders. CC and CC-k variants show no significant predictive power at any order, confirming that unweighted combinatorial metrics are blind to coverage gaps when failure density is low and operationally concentrated.

AsFault. The seven-feature, near-saturated regime produces no significant PWCC correlations at either interaction order. The omitted regions contain failures (113–472 per split), but PWCC already scores near 1.0 for the retained $\mathcal{D}_{\text{test}}$ — leaving no leverage to track the gap. cov_{IS} achieves r_s = +0.867* at t=3, and area_{IS} achieves r_s = +0.709* at t=2; both reflect the size of $\mathcal{D}_{\text{valid}}$ rather than genuine coverage gap detection, and neither provides a stopping criterion.

Frost. At t=2 and t=3, no metric achieves significance: the structural profiling–testing mismatch (86.7% dry-weather profiling) suppresses $\tilde{P}$ for the omitted adverse conditions, removing the signal that PWCC relies on. At t=4, CC-k3 (r_s = +0.954*), CC-k2 (r_s = +0.922*), CC_{t4} (r_s = +0.858*), and $\tilde{P}$ (r_s = +0.858*) appear significant. This is a confound: larger $|\mathcal{D}_{\text{valid}}|$ simultaneously produces higher failure counts and higher combinatorial coverage of the omitted region (r_s(n_valid, failure_count) = +0.890*). PWCC_{t4} ($r_s = -0.629$, $p = 0.051$) resists this confound because its exposure-based sufficiency criterion requires adequate test duration, not merely any contact with a combination.

RQ2

$\tilde{P}_{\text{omit}}$ is the strongest predictor of failure concentration (DeepCollision: r_s = + 0.927*; DeepScenario: r_s = + 0.799*), validating PWCC's probability-weighting design. PWCC correctly signals gaps on DeepCollision (r_s = − 0.830*); PWCC-uniform produces the opposite sign (r_s = + 0.818*), certifying adequacy where the system is most exposed. CC and CC-k show no significant signal on DeepCollision or DeepScenario. The CC-k Frost t=4 correlations are a size confound that PWCC's exposure threshold resists.

5.3 RQ3: False Sufficiency Under ODD Omission

Table 5 reports mean FSR and UFC across ten omission splits per ($dataset$, t) cell. EFC is excluded: it never reached combinatorial saturation on any split and therefore issued no adequacy verdict, producing FSR = UFC = 0 trivially. PWCC-Uniform and PWCC-Empirical are included as ablations to isolate the contribution of probability weighting.

Headline result. PWCC substantially reduces false sufficiency relative to CC, most clearly on DeepScenario and in UFC. On DeepScenario, PWCC attains FSR = 0 and UFC = 0 across all interaction orders and all ten splits — it never issues a false adequacy verdict. CC, by contrast, reaches mean UFC ≥ 99.7% at $t \in \{2, 3\}$ on the same dataset: when CC does certify adequacy, it conceals virtually *all* unseen failures. This complete elimination of false sufficiency directly validates the claim that probability-weighted exposure is the decisive factor for safe adequacy assessment.

AsFault (t = 2). PWCC's mean FSR of 0.24% is 1.8× lower than CC (0.43%), and its mean UFC of 8.2% is 3.9× lower than CC (32.1%). The low absolute values reflect the near-saturated 7-feature space: most splits are already well covered, leaving limited room for false verdicts. At t = 3 the gap widens: PWCC mean UFC of 20.6% versus CC's 70.6%, a 3.4× reduction. PWCC-Uniform (UFC = 28.3%) and PWCC-Empirical (UFC = 22.6%) both fall between PWCC and CC, confirming that the BN-learned probability weighting — not merely the exposure threshold — drives the improvement.

DeepCollision ($t \in \{2, 3\}$). PWCC's mean FSR of 0.86% at t = 2 is 2.8× lower than CC (2.41%), and its UFC of 6.4% is 4.9× lower than CC (31.2%). At t = 3 the UFC gap is 2× (PWCC 19.4% vs. CC 36.2%). PWCC-Uniform produces UFC = 0 at t = 2 and 4.7% at t = 3, an artefact of its near-zero sufficiency rate: uniform weighting declares almost no region sufficient, suppressing false positives but providing no useful adequacy signal. At t = 4 all combinatorial metrics converge: the high dimensionality forces full coverage of every evaluated split, leaving no PWCC–CC differential.

DeepScenario ($t \in \{2, 3, 4\}$). This is the starkest result. PWCC issues *zero* false adequacy verdicts across all 30 evaluated splits (FSR = 0, UFC = 0). CC systematically fails: at t = 2 it certifies adequacy in every split while concealing 100% of unseen failures (UFC = 100.0% in 9 of 10 splits); at t = 3 it reaches mean UFC = 99.7%. The mechanism is structural: DeepScenario's 1.7% failure rate means that all pairwise combinations are trivially covered by the test suite, so CC always saturates and always certifies, regardless of whether the omitted region contains failures. PWCC withholds its verdict because the BN assigns low probability mass to the sparsely observed omitted combinations, correctly requiring more exposure before certifying. PWCC-Uniform mirrors CC's pathology (UFC = 0 only

Table 4: RQ2: Spearman r_s between metric score on $\mathcal{D}_{\text{test}}$ and failure count in $\mathcal{D}_{\text{valid}}$ (10 ODD omission splits per dataset). †: $p>0.05$; bold: highest $|r_s|$ in the correct direction per dataset. AsFault supports $t\leq 3$ only (7 features). PWCC-uni at t=4 on DeepCollision is N/A (constant score under uniform weighting).

	AsFault		DeepCollision			DeepScenario			Frost		
Metric	t=2	t=3	t=2	t=3	t=4	t=2	t=3	t=4	t=2	t=3	t=4
PWCC	$-0.503^{\dagger}$	$+0.139^{\dagger}$	**−0.830**	−0.661	$-0.455^{\dagger}$	$+0.026^{\dagger}$	$+0.543^{\dagger}$	−0.671	$+0.320^{\dagger}$	$-0.013^{\dagger}$	$-0.629^{\dagger}$
CC	$+0.261^{\dagger}$	$+0.455^{\dagger}$	$+0.263^{\dagger}$	$+0.406^{\dagger}$	$+0.152^{\dagger}$	$+0.296^{\dagger}$	$+0.098^{\dagger}$	$-0.191^{\dagger}$	$-0.031^{\dagger}$	$-0.312^{\dagger}$	+0.858
EFC	+0.636	$-0.018^{\dagger}$	$-0.079^{\dagger}$	$+0.273^{\dagger}$	$-0.467^{\dagger}$	$+0.567^{\dagger}$	$-0.189^{\dagger}$	$-0.043^{\dagger}$	$-0.037^{\dagger}$	$+0.070^{\dagger}$	$-0.025^{\dagger}$
$\tilde{P}$	$+0.527^{\dagger}$	$+0.236^{\dagger}$	**+0.927**	+0.770	+0.806	+0.683	**+0.799**	$+0.332^{\dagger}$	$-0.314^{\dagger}$	$-0.070^{\dagger}$	+0.858
PWCC-uni	−0.794	$+0.261^{\dagger}$	+0.818	$+0.067^{\dagger}$	N/A	$+0.291^{\dagger}$	$+0.396^{\dagger}$	$-0.369^{\dagger}$	$+0.140^{\dagger}$	$+0.412^{\dagger}$	$-0.203^{\dagger}$
PWCC-emp	$-0.491^{\dagger}$	$+0.067^{\dagger}$	−0.867	$-0.564^{\dagger}$	$-0.624^{\dagger}$	$-0.139^{\dagger}$	+0.701	$-0.616^{\dagger}$	$+0.108^{\dagger}$	$-0.057^{\dagger}$	$-0.572^{\dagger}$
CC-k2	$+0.243^{\dagger}$	$+0.382^{\dagger}$	$+0.462^{\dagger}$	$+0.321^{\dagger}$	$+0.515^{\dagger}$	$-0.138^{\dagger}$	$+0.156^{\dagger}$	$-0.062^{\dagger}$	$-0.024^{\dagger}$	$+0.070^{\dagger}$	+0.922
CC-k3	$+0.006^{\dagger}$	$+0.333^{\dagger}$	$+0.529^{\dagger}$	$+0.236^{\dagger}$	$+0.515^{\dagger}$	$-0.172^{\dagger}$	$-0.067^{\dagger}$	$-0.308^{\dagger}$	$+0.118^{\dagger}$	$+0.338^{\dagger}$	**+0.954**‡
area$_{IS}$	+0.709	$-0.006^{\dagger}$	$+0.430^{\dagger}$	$-0.309^{\dagger}$	$-0.067^{\dagger}$	$-0.329^{\dagger}$	$+0.006^{\dagger}$	$-0.142^{\dagger}$	$-0.455^{\dagger}$	$+0.108^{\dagger}$	+0.826
area$_{bugs}$	$+0.285^{\dagger}$	$-0.249^{\dagger}$	$+0.576^{\dagger}$	$+0.030^{\dagger}$	$+0.067^{\dagger}$	$-0.329^{\dagger}$	$+0.006^{\dagger}$	$-0.142^{\dagger}$	$-0.332^{\dagger}$	$+0.025^{\dagger}$	+0.737
cov$_{IS}$	$+0.018^{\dagger}$	**+0.867**	$-0.479^{\dagger}$	−0.806	$-0.273^{\dagger}$	$-0.342^{\dagger}$	$-0.463^{\dagger}$	$-0.049^{\dagger}$	$+0.025^{\dagger}$	$+0.057^{\dagger}$	$+0.534^{\dagger}$

‡ Confound: r_s(n_valid, failure_count) = +0.890*; CC-k Frost t=4 correlations are driven by held-out set size, not coverage signal — see text. Horizontal rules separate PWCC/baselines, ablations, and TISA.

Table 5: RQ3: Mean FSR (%) and UFC (%) across ten ODD omission splits. Lower is better for both. Entries marked "—" indicate no adequacy verdict was issued (metric did not saturate). †: at t = 4 all metrics converge on DeepCollision due to full coverage saturation. AsFault supports $t \leq 3$ only.

		AsFault		DeepCollision		DeepScenario		Frost	
Metric	t	FSR	UFC	FSR	UFC	FSR	UFC	FSR	UFC
PWCC	2	**0.24**	**8.2**	**0.86**	**6.4**	**0.00**	**0.0**	**1.98**	43.6
	3	**4.64**	**20.6**	**6.50**	**19.4**	**0.00**	**0.0**	12.27	**24.0**
	4	—	—	†19.75	†11.5	**0.00**	**0.0**	**1.09**	**10.0**
CC	2	0.43	32.1	2.41	31.2	0.72	100.0	2.05	87.9
	3	9.17	70.6	14.64	36.2	1.27	99.7	2.12	94.3
	4	—	—	†20.03	†11.9	1.09	95.8	—	—
CC-k2	2	0.34	22.7	1.92	16.7	0.71	99.2	2.05	88.0
	3	7.65	54.9	7.17	23.9	1.28	99.4	2.10	93.5
	4	—	—	15.39	5.0	1.13	92.7	—	—
CC-k3	2	0.21	13.1	0.86	9.2	0.71	98.2	2.05	88.0
	3	6.06	42.8	5.77	13.9	1.29	99.1	2.05	88.6
	4	—	—	5.85	0.8	1.12	83.3	—	—
PWCC-Uni.	2	**0.00**	**0.0**	**0.00**	**0.0**	—	—	2.03	97.0
	3	5.39	28.3	2.39	4.7	—	—	2.17	98.9
	4	—	—	†19.84	†11.9	2.10	11.5	0.74	23.8
PWCC-Emp.	2	0.47	16.3	0.90	5.2	—	—	1.98	42.8
	3	4.98	22.6	6.33	18.9	—	—	12.31	24.0
	4	—	—	†20.04	†11.5	—	—	1.09	10.0

because it never certifies), confirming that uniform weighting cannot distinguish safe from unsafe regions in low-failure-rate settings.

Frost ($t \in \{2, 3\}$). Frost presents a structural mismatch: the profiling distribution is dominated by dry clear-weather conditions (86.7% dry), while omitted regions are drawn uniformly and frequently target adverse weather. At t = 2, PWCC and CC produce near-identical FSR ($\approx$ 2%) and similar UFC (PWCC 43.6% vs. CC 87.9%) — a 2× advantage for PWCC. At t = 3, PWCC-Empirical and PWCC (both UFC 24.0%) substantially outperform CC (UFC 94.3%). The modest PWCC advantage relative to DeepScenario reflects the mismatch: the BN correctly assigns low probability to adverse conditions, but those are precisely the omitted regions, so PWCC can still certify the retained clear-weather majority as adequate. This is a structurally coherent outcome, not a failure of the metric, and underscores the importance of verifying $\mathcal{D}_{\text{prof}}$ alignment with the intended deployment distribution before drawing adequacy conclusions (see Section 6).

Ablation: the role of probability weighting. PWCC-Uniform attains FSR = UFC = 0 at t = 2 on AsFault and DeepCollision not by detecting coverage gaps but by rarely certifying any region adequate — *conservative failure*: it avoids false positives only by almost never issuing a verdict, carrying no adequacy signal. On Frost it is worse than CC (UFC = 97.0% at t = 2): without operational weighting it certifies the dominant dry-weather regions while concealing nearly all failures in the adverse-weather omissions. PWCC-Empirical tracks PWCC at low interaction orders but degrades at t = 3, where sparse profiling data makes empirical frequency counts unreliable. The BN-learned distribution is the source of PWCC's advantage.

RQ3

PWCC is the only metric that suppresses false sufficiency when the operational distribution carries signal: FSR = UFC = 0 on DeepScenario across all 30 splits vs. CC's mean UFC ≥ 95.8%; UFC 2–5× lower than CC on AsFault and DeepCollision. Ablation confirms that BN-learned probability weighting is the decisive factor — PWCC-Uniform produces spurious zeros through under-certification, and on Frost it is worse than CC.

5.4 RQ4: Stopping Behaviour

RQ1–RQ3 characterise the metrics at a fixed suite size; RQ4 complements them by evaluating each metric at the point it declares adequacy. Each metric triggers its own stop; at that point we record the fraction of the campaign consumed (*Used*) and the fraction of distinct failure-triggering t-way combinations surfaced (*Cap*). Table 6 reports mean±std over ten orderings.

The decisive case: sparse failures. On DeepScenario (1.7% failures), CC's plateau triggers at just 6.4% of the campaign having surfaced only 54.6% of failure-combinations — it declares adequacy with 198 of 437 still undiscovered; EFC is worse, stopping at 1% with 79% concealed. PWCC withholds its verdict until 96.3% and surfaces 99.6%, leaving two combinations undiscovered. The pattern holds at every order and on Frost, the other mismatch dataset (Cap 66.1% vs. CC's 19.7% at t=2): wherever operational signal is sparse, combinatorial saturation stops early and conceals failures — the false-sufficiency mechanism of RQ3, now in stopping terms.

Failure-dense regimes: efficient, risk-calibrated stops. Where failures are dense, PWCC stops *early* rather than late. On AsFault at t=2 it surfaces 83.4% of failure-combinations having consumed only 30.5% of the campaign — half of CC's 61.2%. On DeepCollision at t=2 it leaves 3× fewer combinations undiscovered than CC at the stop (57 vs. 152). Where PWCC surfaces a smaller share of raw combinations than an exhaustive-saturation baseline (AsFault t=2; DeepCollision t=3), the residual mass is operationally negligible: PWCC halts when the frequency-weighted operational gap closes, not when every combination is mechanically covered — the saturation regime of RQ1 seen from the stopping side.

RQ4

Judged at its own stopping point, PWCC surfaces 99.6% of failure-triggering combinations on DeepScenario while CC stops after 6.4% of the campaign with 45% still concealed — false sufficiency in stopping terms. On failure-dense data PWCC stops early and efficiently (83% of failure modes at 30% of the AsFault campaign; 3× fewer undiscovered than CC on DeepCollision), grounding the stop in risk-calibrated operational exposure rather than combinatorial saturation.

5.5 RQ5: Sensitivity, Stability, and Efficiency

Monotone growth and stability. Figure 3 plots mean PWCC score and min–max band across ten repetitions as suite size increases from 500 to 5,000 scenarios. The score grows strictly monotonically across all $11 \times 10 \times 10 = 1{,}100$ evaluated (dataset, t, size, rep) combinations — 11 (dataset, t) pairs, since AsFault supports $t \leq 3$ — no non-monotone step was observed. Score variance shrinks sharply with size: standard deviation falls below 0.006 at $n \geq 2{,}500$ on both AsFault and DeepCollision, confirming that PWCC provides consistent, low-noise feedback throughout a test campaign. The dashed red line marks the $\varepsilon = 0.05$ adequacy threshold; crossing it defines the earliest stopping point under the baseline parameters. The dataset-dependent growth shape is itself informative: Frost saturates within 1,000 scenarios at $t = 2$ due to its compact operational distribution; AsFault requires ~2,000; DeepCollision ~3,000; and DeepScenario never certifies at $t = 2$ under the baseline risk budget — the correct behaviour for a 1.7% failure-rate dataset (see RQ3).

Risk budget ρ, confidence level α, and interaction order t. Table 7 reports mean PWCC score for the $\rho \times \alpha$ grid at $t = 2$ on DeepCollision. Both parameters have directional, interpretable effects: lower ρ or lower α tightens $E_{\min}(c)$, reducing covered mass and score. Sensitivity to interaction order t is consistent with ρ: higher t lowers the mean score as more combinations require sufficient exposure, but the directional response to ρ is preserved across all

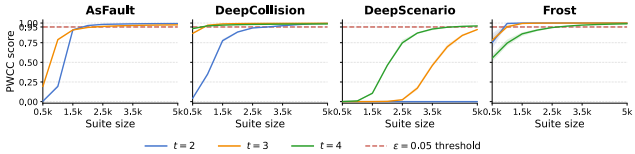


Figure 3: PWCC score vs. suite size for all four datasets and $t \in \{2, 3, 4\}$ (AsFault: $t \leq 3$). Bold lines: mean over 10 repetitions; bands: min–max. Dashed red: $\varepsilon = 0.05$ threshold. Score grows strictly monotonically across all 1,100 configurations.

orders. Sensitivity saturates as exposure accumulates — at $n = 3{,}000$ the gap between the strictest and most lenient ρ is under 0.02 on DeepCollision. Since ρ, α, and t are user-defined requirements, not free hyperparameters, their effect on the score is precisely the intended behaviour of a risk-informed metric.

Coverage threshold ε. ε affects only the binary adequacy verdict, not the continuous PWCC score: values are identical across all $\varepsilon \in \{0.01, 0.05, 0.10, 0.20\}$ at every size and repetition on all datasets. This cleanly separates the two roles of the framework: the score measures ODD coverage; ε encodes the minimum fraction required to stop — a policy decision independent of the coverage signal itself.

Monte Carlo sample size N and discretisation. Score variation across a 10× increase in N (10,000 to 100,000) is at most 0.006 on all four datasets, confirming that $N = 50{,}000$ is well past the point of diminishing returns. To assess sensitivity to the discretisation breakpoints, we repeated the RQ3 evaluation on DeepCollision at $t = 2$ using 0.5× and 2× the number of bins per feature. Mean FSR changed by at most 0.3 percentage points and mean UFC by at most 1.1 percentage points across both perturbations, confirming that adequacy verdicts are stable with respect to moderate discretisation variation.

Computational efficiency. Table 8 reports evaluation time on the full datasets. At $t = 2$, PWCC completes in under 70 seconds across all datasets, faster than TISA on every dataset. At $t = 3$, cost ranges from 18 seconds (AsFault) to 6.6 minutes (DeepCollision). At $t = 4$, evaluation remains under 3 minutes except on DeepCollision, where 18 features produce 499,316 combinations and evaluation takes 22 minutes — a known limitation of combinatorial methods at high interaction orders and feature dimensionality. The combination-level independence of PWCC's sufficiency criterion directly supports parallelisation for time-sensitive settings.

6 Threats to Validity

Internal validity. The homogeneous Poisson model treats critical events under a combination as an *arrival count across the campaign*, not a causal chain within any single failure — a standard reliability-engineering device for estimating the exposure needed to detect a fault at rate $\lambda_{\max}(c)$ with confidence $1 - \alpha$. It ignores temporal clustering of failures; on high-failure-rate data this is conservative, demanding more exposure than strictly necessary — the safe direction for a stopping criterion. The credited-mass construction (Equation 2) splits each scenario's *own* mass equally across *its* $K(s)$ valid t-way projections; it does not assign equal mass to all combinations, so frequent combinations accumulate more and rarity is preserved. Equal within-scenario attribution is the minimal

Table 6: RQ4: Stopping behaviour. *Used*: % of the campaign consumed at the metric's stopping point; *Cap*: % of distinct failure-triggering t-way combinations surfaced at that point. Mean±std over ten random orderings; higher *Cap* at lower *Used* is better. Bold: highest *Cap* per (dataset, t). AsFault supports $t \leq 3$.

		AsFault		DeepColl.		DeepScen.		Frost	
Metric	t	Used	Cap	Used	Cap	Used	Cap	Used	Cap
PWCC	2	30.5±0.7	83.4±1.6	34.7±0.8	**97.3**±0.8	96.3±0.7	**99.6**±0.9	12.7±0.9	**66.1**±7.7
	3	82.2±3.5	95.3±1.0	12.5±0.7	81.4±1.7	66.1±3.0	**91.2**±1.2	21.7±2.5	40.3±6.4
	4	—	—	74.0±1.9	**96.4**±0.4	77.0±2.0	**90.9**±1.7	94.1±1.4	**95.2**±1.5
CC	2	61.2±5.9	93.3±0.9	12.0±2.3	92.8±0.8	6.4±1.1	54.6±7.4	3.0±0.0	19.7±6.2
	3	77.4±1.9	94.0±0.6	33.0±3.6	91.7±0.4	20.6±2.9	61.8±4.7	16.3±0.5	31.4±3.5
	4	—	—	50.4±2.1	91.3±0.3	41.3±2.2	70.2±2.7	73.8±0.8	79.3±1.4
CC-k2	2	65.8±3.0	94.0±0.6	21.4±3.6	95.5±0.5	12.5±2.6	69.2±5.1	4.2±0.4	27.8±5.4
	3	81.6±2.2	95.3±0.5	48.6±3.4	95.1±0.4	35.1±1.3	75.8±3.1	25.7±0.5	45.7±3.3
	4	—	—	66.4±1.3	94.9±0.3	55.0±3.1	80.6±3.1	89.5±0.5	91.7±1.4
CC-k3	2	73.0±2.4	**95.4**±0.8	27.0±3.8	96.5±0.5	17.3±2.5	74.7±3.7	5.9±0.3	39.1±7.0
	3	83.2±1.3	**95.7**±0.6	55.3±2.4	**96.0**±0.4	44.6±3.0	81.8±3.5	34.1±0.6	**55.7**±2.5
	4	—	—	72.7±0.5	**96.2**±0.3	62.8±2.5	84.5±2.9	94.5±0.5	**95.3**±0.9
EFC	2	3.2±0.4	48.2±3.2	1.0±0.0	65.5±3.6	1.0±0.0	21.2±9.2	2.0±0.0	14.1±7.1
	3	13.7±0.8	54.8±1.9	2.2±0.4	54.4±3.7	1.4±0.5	12.0±5.3	10.0±0.5	19.7±3.7
	4	—	—	6.0±0.0	54.6±1.4	3.9±0.3	14.8±5.1	45.9±1.3	54.2±2.7

Table 7: RQ5: Mean PWCC score over all suite sizes and repetitions for the $\rho \times \alpha$ grid ($t = 2$, DeepCollision). Baseline: $\rho = 0.01$, $\alpha = 0.05$ (bold).

$\rho \setminus \alpha$	0.01	0.05	0.10	0.20
0.001	0.001	0.010	0.026	0.092
0.005	0.342	0.556	0.658	0.771
0.010	0.658	**0.787**	0.850	0.908
0.050	0.976	0.992	0.994	0.998
0.100	0.994	0.998	0.999	0.999

Table 8: Evaluation time (seconds) on full datasets.

Dataset	$t = 2$	$t = 3$	$t = 4$	TISA
AsFault	9.9	17.5	—	47.4
DeepScenario	12.3	104.6	177.2	98.2
Frost	14.8	63.8	83.2	45.9
DeepCollision	66.8	398.6	1327.3	107.0

assumption absent a justified importance model; severity- or causal-contribution weighting is compatible and left to future work, and the effect of alternative schemes in highly dependent feature spaces warrants further study. Likewise, the quartile-anchored breakpoints are risk-agnostic; risk-aware binning that isolates safety-critical value ranges is a compatible refinement, and verdicts are stable under moderate bin-count variation (RQ5).

External validity. Results span four datasets covering distinct failure regimes (high/low failure rate, profiling–testing mismatch, near-saturation), but all are autonomous driving benchmarks. Generalisation to other cyber-physical domains requires further validation. Failure severity is treated uniformly; integrating consequence weights into $\lambda_{\max}(c)$ is an important direction for domains where hazard magnitude varies substantially across conditions.

Construct validity. The per-combination budget ρ does not bound aggregate risk: the global sum $\sum_c \tilde{P}(c) \cdot \lambda(c)$ can reach $\rho \cdot |C_t|$, which grows with feature space size. Practitioners should treat ρ as a per-combination safety requirement and use the coverage threshold ε as the primary global control; by Proposition 3.2, when $\text{Uncovered}_t(\mathcal{T}) < \varepsilon$, the operational mass exposed to uncontrolled risk is provably bounded by ε.

PWCC's guarantee is conditional on $\mathcal{D}_{\text{prof}}$: representativeness of the profiling data is an explicit assumption, not a derived property. A profiling–deployment mismatch (as in Frost) suppresses $\tilde{P}$ for underrepresented conditions, reducing sensitivity to gaps there; the BDeu prior keeps $\tilde{P}(c) > 0$ for unobserved cells but does not correct profiling *bias*. Practitioners should therefore verify that key per-feature marginals of $\mathcal{D}_{\text{prof}}$ match the intended deployment distribution before interpreting a verdict. No uncertainty from BN parameter estimation is propagated into $\tilde{P}(c)$ or adequacy decisions; near the ε threshold, credible intervals on $\tilde{P}(c)$, for which conformal prediction (Section 2) is a natural vehicle, would strengthen certification arguments. The empirical sensitivity of these modelling choices is quantified directly: the PWCC-Uniform and PWCC-Empirical ablations (RQ3) isolate the effect of the weighting model, and the N and discretisation analyses (RQ5) show adequacy verdicts are stable to sampling budget and binning.

7 Conclusion

Scenario-based testing of autonomous systems has lacked a principled answer to its most consequential question: when is enough enough? Existing metrics measure what has been tested; none quantify how much of what the system will actually encounter remains insufficiently exercised.

PWCC fills this gap by grounding combinatorial coverage in a learned operational distribution and requiring combination-specific exposure calibrated to explicit risk and confidence parameters. When the stopping criterion is met, at most a fraction ε of expected operational encounters involve insufficiently tested conditions — a bound on residual untested operational coverage directly usable within a SOTIF safety argument.

Empirically, probability weighting is decisive: without it, combinatorial metrics certify adequacy while concealing virtually all unseen failures. With it, PWCC achieves FSR = UFC = 0 on DeepScenario, where CC conceals ≥95.8% of unseen failures, and reduces UFC by up to 4.9× on failure-rich data. The score is stable, monotonic, and evaluates in under 70 seconds at $t = 2$ across all datasets.

Future work. Three directions are most pressing. First, *uncertainty quantification*: propagating BN parameter uncertainty into credible intervals on $\tilde{P}(c)$ near the ε threshold would strengthen adequacy arguments under profiling–deployment mismatch. Second, *severity integration*: extending $\lambda_{\max}(c)$ to incorporate consequence weights would refine the risk allocation. Third, *adaptive test generation*: using the uncovered mass to guide scenario selection toward insufficiently tested combinations would close the loop between adequacy measurement and test campaign planning.

Data Availability Statement

The PWCC implementation, all datasets, and evaluation scripts are available at https://doi.org/10.5281/zenodo.19181612.

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